Interpretable GATs for Expected Goals Prediction

February 6, 2026

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1. Introduction

Expected Goals (xG) is a fundamental metric in soccer analytics, yet it typically evaluates only the final shot, overlooking the complex possession chain that created the opportunity. This work addresses this limitation by modeling offensive possessions using Graph Attention Networks (GATs) (Veličković et al., 2018), where players and passes form the graph structure. To overcome the opacity of standard attention scores, I introduce a modified attention mechanism that explicitly incorporates spatial priors specifically pass length and goal proximity. This approach uses xG prediction as a supervisory signal to highlight the specific progressive actions most critical to generating high-quality scoring chances, offering match analysts a transparent tool to dissect the anatomy of successful attacks.

2. Related Work

Expected Goals (xG) revolutionized match analysis by quantifying the probability of a shot resulting in a goal. However, traditional xG models have significant limitations; they often overlook contextual variables such as weather conditions, defensive positioning, player biomechanics (e.g., preferred foot), and player roles. Furthermore, because xG focuses exclusively on the final shot, it fails to evaluate the possession chain leading up to the event. Consequently, it cannot identify which players contributed most effectively to creating the scoring opportunity.

Expected Threat (xT) (Singh, 2018) addresses this limitation by introducing a probabilistic model that evaluates non-shooting actions. Unlike xG, xT credits players for moving the ball into "dangerous" zones that increase the likelihood of a goal. However, similar to early xG models, xT is primarily location-based and is often limited by its reliance on ball position coordinates rather than the full context of play.

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Deep Learning and Applied AI 2025, Sapienza University of Rome, 2nd semester a.y. 2024/2025.

On-Ball Value (OBV) (Dorrington, 2025) extends the concepts introduced by Singh by incorporating a broader range of game states. By analyzing events such as ball recoveries and specific action types, OBV provides a more comprehensive framework for evaluating a player's total contribution to a possession.

3. Method

3.1. Baselines

For the model architecture, I adopted a Graph Attention Network (GAT) (Veličković et al., 2018). This choice allows the model to explicitly learn how information flows between players by assigning attention weights to the edges of the graph, which represent passes between players during a possession. By leveraging these attention mechanisms, the network can weigh different passes differently when predicting the expected goals (xG) value of the final shot. This not only improves the model's expressive power when capturing complex interactions among players, but also provides a degree of interpretability, as the learned attention scores can be analyzed to understand the relative importance of individual passes and player interactions within each possession.

3.2. Contribution

Building upon the baseline model, I enhanced its interpretability by modifying the attention mechanism originally proposed in the Graph Attention Network (GAT) framework (Veličković et al., 2018). Specifically, I introduce a novel, adaptive attention formulation that explicitly incorporates the spatial context of each pass: its magnitude (length) and its directionality relative to the opposing goal.

The standard attention mechanism is modulated by two scalar coefficients, w_1 and w_2, which weight the attention logits based on spatial features. The modified attention coefficients $\alpha_{ij}^{(h)}$ are computed as follows:

$$\alpha_{ij}^{(h)} = \frac{\exp\left(e_{ij}^{(h)} \cdot w_1^{(h)} \cdot w_2^{(h)}\right)}{\sum_{k\in\mathcal{N}(i)} \exp\left(e_{ik}^{(h)} \cdot w_1^{(h)} \cdot w_2^{(h)}\right)} \tag{1}$$

Here, $e_{ij}^{(h)}$ represents the unnormalized attention scores. The weighting coefficients are defined as follows:

3.2.1. Pass Length Reward (w_1)

$$w_1^{(h)} = \exp\left(\lambda \cdot \|\mathbf{c}_j^{(h)} - \mathbf{c}_i^{(h)}\|_2\right) \tag{2}$$

where $\mathbf{c}_i$ and $\mathbf{c}_j$ denote the spatial coordinates of the sender and receiver, respectively. The term w_1 serves as a prioritization factor for pass length; as the Euclidean distance between nodes increases, the attention weight scales up.

3.2.2. Goal Proximity Reward (w_2)

$$w_2^{(h)} = \exp\left(-(1-\lambda) \cdot \|\mathbf{c}_j^{(h)} - \mathbf{g}\|_2\right) \tag{3}$$

where $\mathbf{g}$ represents the fixed coordinates of the attacking goal. This coefficient evaluates the progressive nature of the pass. As the receiver's distance to the goal decreases, the weight increases. Conversely, passes ending far from the goal are penalized. The hyperparameter $\lambda \in [0, 1]$ controls the balance between these two spatial incentives.

Under this modified attention mechanism, forward and progressive passes such as long balls played directly from the defensive line toward the attacking goal are assigned higher attention weights, reflecting their greater potential impact on the resulting shot. In contrast, backward or non-progressive passes are penalized with lower attention scores. This design encourages the model to focus on passes that are influential in advancing play and generating dangerous situations, enhancing both predictive performance and model transparency.

4. Experiments and Results

4.1. Data Preprocessing

The dataset used for this project is the StatsBomb open data (StatsBomb), which contains detailed event-level information from a large number of men's and women's soccer matches. For each ball possession that ended with a shot, I constructed a graph in which players are represented as nodes and passes as edges. Each graph is associated with the expected goals (xG) value of the final shot that concluded the possession.

4.2. Experiment 1 – xG Prediction

The first experiment focuses on predicting the expected goals (xG) value from the pass graph. This task is relatively straightforward, as xG is typically computed using primarily the shot location and a small number of additional, simple contextual features. As expected, the model converges quickly: as reported in Table 1, even with only five training epochs, it achieves a very low validation loss.

Table 1. Validation set metrics for the xG prediction task.

Metric	Value
Mean Squared Error (MSE)	0.007344
Root Mean Squared Error	0.085698
Mean Absolute Error	0.057160
Median Absolute Error	0.044162
R^2 Score	0.033714
Pearson Correlation	0.188143
Spearman Correlation	0.211998
Explained Variance	0.033983
Max Error	0.887327
Mean Squared Log Error	1.073742

4.3. Experiment 2 – Interpretable Attention Scores

The second experiment aims to build an interpretable model capable of identifying the most influential pass within a ball possession that leads to a given xG value. To this end, I analyze the modified attention mechanism applied to the edges of the pass graph and select, for each possession, the pass associated with the highest attention score.

As discussed in (Lipton, 2017), interpretability should be treated with caution: a model cannot be deemed interpretable solely on the basis of a few cherry-picked examples. For this reason, I avoid presenting individual qualitative examples as evidence of interpretability. Instead, I adopt an aggregate analysis by constructing a heatmap in which each pixel's intensity reflects the frequency with which a pass traverses that spatial region.

Specifically, I generate one heatmap using only the highest-attention pass from each ball possession in the validation set, and compare it against a baseline heatmap built from all passes in the same validation set. This comparison allows me to assess whether the attention mechanism consistently highlights spatial patterns that differ meaningfully from the overall pass distribution.

5. Conclusion

This study demonstrates that GATs can effectively model the relationship between passing sequences and xG. By modifying the attention mechanism to reward spatial progression, the model aligns its focus with tactical intuition, specifically prioritizing forward and dangerous passes. Aggregate spatial analysis confirms that these interpretable attention weights provide meaningful insights into chance creation without sacrificing predictive performance.

References

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Appendix

Interpretability Analysis

Figure 1 illustrates the spatial density of the highest-attention pass selected for each possession, whereas Figure 2 reports the overall distribution of all passes across the pitch.

A clear difference emerges between the two representations. While the pass density map highlights regions of frequent ball circulation primarily in the midfield and build-up zones the attention density map concentrates predominantly in the final third of the pitch. This indicates that the model has learned to assign higher importance to passes occurring in areas that are more likely to generate dangerous situations, such as assists or opportunities that destabilize the opponent's defensive structure.

This behavior is consistent with tactical intuition in football, where decisive actions tend to occur closer to the opponent's goal rather than in zones of high pass volume. Although isolated high-attention passes can occasionally be found in other areas of the pitch (e.g., through balls or progressive passes from deeper positions), the overall spatial pattern suggests that the model is not simply reproducing pass frequency, but instead capturing context-dependent relevance.

Overall, this analysis provides qualitative evidence that the attention mechanism contributes to the interpretability of the model by highlighting passes that are both spatially and tactically meaningful, supporting the validity of the learned representations.

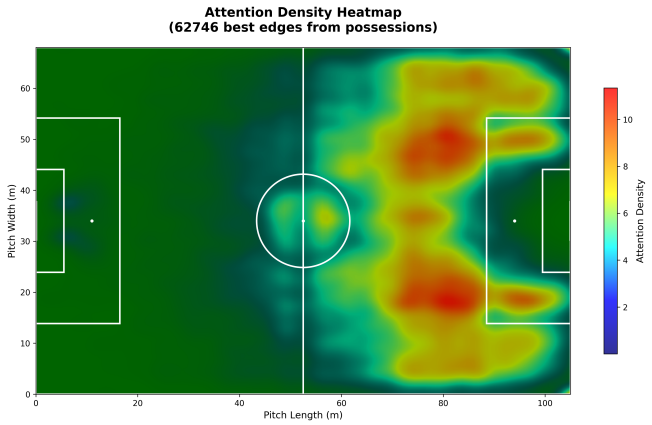

Figure 1. Attention density heatmap (highest-attention pass per possession).

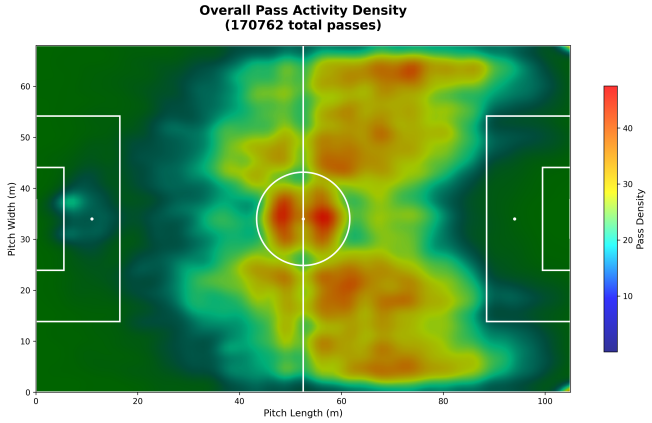

Figure 2. Pass density heatmap (all passes in the validation set).

Other Experiments

At an early stage, this project was designed around a progressive Graph Attention Network (GAT), where the graph was incrementally constructed to represent the temporal evolution of a possession. The underlying idea was that progressively revealing interactions would help the model better capture the buildup dynamics leading to dangerous situations.

However, empirical evaluation showed that this approach did not provide a measurable performance advantage compared to using a single full graph representing the entire possession. In practice, the progressive formulation introduced additional complexity without improving either predictive performance or interpretability.

For this reason, the focus of the project was shifted away from graph construction strategies and toward a deeper analysis of the attention mechanisms themselves, which proved to be more informative in highlighting the most relevant interactions within a possession.